



Figure 2: The proposed pipeline for generating instruction-violating responses for Task 2.

Model	EN	FR	DE	ES	ZH	AR	HI
Claude3.7-Sonnet	0.7608/0.7298	0.7558/0.7233	0.7533/0.7260	0.7490/0.7283	0.7407/0.7170	0.7510/0.7205	0.7472/0.7152
Claude3.5-Sonnet-V2	0.7215/0.6874	0.7253/0.6776	0.7187/0.6763	0.7207/0.6887	0.7175/0.6640	0.7114/0.6582	0.7033/0.6524
Claude3.5-Sonnet-V1	0.6847/0.6854	0.6867/0.6811	0.6869/0.6784	0.6834/0.6796	0.6615/0.6604	0.6483/0.6551	0.6435/0.6496
Llama3.3-70B	0.7283/0.5636	0.7273/0.5346	0.7303/0.5388	0.7250/0.5356	0.6978/0.5545	0.6266/0.4970	0.6405/0.5129
Qwen3-14B	<u>0.7046/0.6937</u>	0.6968/0.6905	0.6958/0.6872	0.6965/0.6847	0.6965/0.6907	0.6920/0.6837	0.6895/0.6758
Gemma3-27B-IT	<u>0.6968/0.5616</u>	0.6890/0.5436	0.6773/0.5477	0.6935/0.5482	0.6890/0.5404	0.6791/0.5376	0.6804/0.5409

Table 3: Task 2 accuracy comparison between Approach 1/Approach 2 with instruction format f_1 across conversation languages (English (EN), French (FR), German (DE), Spanish (ES), Chinese (ZH), Arabic (AR), Hindi (HI)). The best result per language is **bolded** and the best result per model is underlined.

2) **Instruction Manipulation:** We apply systematic manipulations to both the extracted instruction section and the complete source instruction using an LLM to simulate the following categories of violations:

- **Fine-grained parameter mismatch:** We substitute numerical values (e.g., number of days for which a customer is eligible to return an item) in the extracted section of the instruction with randomly generated numbers, simulating subtle compliance violations.
- **Single-condition action replacement:** For instructions containing one condition with multiple actions, we randomly substitute an original action with either its negation or an alternative action from the same instruction, capturing action-level instruction violations.
- **Multi-condition action replacement:** For instructions with a composition of multiple conditions, we alter the logical condition-action relationships by randomly replacing a target action with an action corresponding to other conditional branches within the same instruction. For example, we could apply standard member return policies for a gold member, violating the true dependence of actions on their corresponding conditions.

3) **Response Generation:** We provide the dialogue history, the manipulated and original instructions, and the ground-truth response to the LLM and prompt it to minimally revise the ground-truth response such that it covers the main difference between the original and manipulated instruction. We explain in the prompt that the generated response must comply with the manipulated instruction. This implicitly guarantees that the generated response is an instruction-violating response.

We used LLM to verify that generated instruction-violating responses do not *entail* ground-truth responses. Of 2288 generated samples, 307 required additional manipulation cycles. Manual verification confirmed 87% accuracy in instruction violations (1982 samples), resulting in a final dataset of 3964 balanced examples (compliant and non-compliant) for Task 2.

Results We compare the six LLMs as binary classifiers for instruction violations in TOD-ProcBench using two approaches:

- **Approach 1:** Given the conversation, next agent response, and relevant instruction section, the LLM should classify the agent’s next response as compliant or instruction-violating.
- **Approach 2:** LLM first attributes the given agent response to the most relevant instruction and then verifies whether the identified instruction and originally provided ground-truth instruction are *entailing* each other. *Entailment* implies a compliant response.

We leverage accuracy as the evaluation metric for Task 2. As Table 3 indicates, most models perform slightly better for English conversations in both approaches. Table 3 also highlights Approach 2’s impact on the compliance evaluation performance as all models demonstrate inferior performance

when relying on instruction-level comparisons. Approach 2 increases the complexity of the compliance evaluation task by having the model first generate the relevant instruction based on the given conversation and then verify compliance with the generated instruction. Table 32 in the Appendix also demonstrates that the LLMs are unable to conduct the compliance evaluation task accurately and have a subtle preference for format f_1 , calling for further studies.

4.3 Task 3: Compliant Response Generation

Inspired by real-world scenarios of conditional generations [Chen et al., 2019, Siddique et al., 2022, Narayan et al., 2023], in Task 3, we investigate ToDs’ conditional generation abilities based on our predefined instructions which require responses to follow the specified guidelines.

Dataset In Task 3, we only examine TODs’ abilities to generate follow-up responses given that the previous conversation is already compliant with given instructions. We use the 3310 partial conversations from the first step of the Task2 pipeline containing instruction-following responses, dialogue histories, and corresponding instructions.

Results We instruct the six LLMs to continue the conversation while following the given instruction. In the prompt, we decompose the problem to identify the steps the agent has followed, select the next most relevant action based on the given instruction, and generate a response that matches the instruction’s retrieved action. We employ Claude3.7-Sonnet (the top-performing Task 2 LLM) as a judge to assess response quality and compliance without ground-truth knowledge. Using its compliance classifications, we calculate compliance rates for Task 3, where higher scores indicate better consistent adherence to instructions.

Model	EN	FR	DE	ES	ZH	AR	HI
Claude3.7-Sonnet	0.9529	0.9553	0.9535	0.9628	0.9616	0.9680	0.9622
Claude3.5-Sonnet-V2	0.9532	0.9505	0.9432	0.9544	0.9529	<u>0.9559</u>	0.9505
Claude3.5-Sonnet-V1	0.9254	0.9353	0.9344	0.9378	0.9375	<u>0.9429</u>	0.9390
Gemma3-27B-IT	<u>0.8952</u>	0.3130	0.3121	0.3009	0.3205	0.3278	0.3323
Llama3.3-70B	0.1435	0.4773	<u>0.5773</u>	0.3565	0.5227	0.3931	0.3834
Qwen3-14B	0.2269	0.2281	0.2254	0.2142	0.2326	0.2369	<u>0.2375</u>

Table 4: Task 3 compliance rate across models and conversation languages (English (EN), French (FR), German (DE), Spanish (ES), Chinese (ZH), Arabic (AR), Hindi (HI)) with instruction format f_1 . The best result per language is **bolded** and the best result per model is underlined.

Table 4 compares the quality of the models’ generations using prior criteria from the given instructions. Larger LLMs tend to exhibit higher compliance in comparison to smaller models, across the conversation language and instruction format dimensions. Some smaller models tend to not respond in the specified response format, leading to many non-compliant responses and very low compliance rates. The performance of all LLMs is not significantly impacted by the language. We observe that the models usually generate longer utterances as it is hard for them to accurately identify one next action from the instruction and to exclusively cover that action. Even though the instruction-following generations may not be semantically close to ground-truth responses, Claude3.7-Sonnet still considers them as compliant.

5 Conclusion

In this work, we apply complex condition-action formatted natural language instructions for ToDs to closely model the complexity of real world complex instructions. With these instruction documents and their application to multilingual conversations, we propose three related tasks: instruction retrieval and next action prediction, instruction-following evaluation, and compliant response generation. We publish benchmarks for each task in TOD-ProcBench and show that advanced LLMs still strive to perform admirably in all of these tasks and with enhanced multilingual capabilities for more challenging, fine-grained tasks. As future works, we plan to extend TOD-ProcBench to additional dataset formats including codified structures across domain-specific instructions.

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References

- Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33, 2020.
- Derek Chen, Howard Chen, Yi Yang, Alexander Lin, and Zhou Yu. Action-based conversations dataset: A corpus for building more in-depth task-oriented dialogue systems. In Kristina Toutanova, Anna Rumshisky, Luke Zettlemoyer, Dilek Hakkani-Tur, Iz Beltagy, Steven Bethard, Ryan Cotterell, Tanmoy Chakraborty, and Yichao Zhou, editors, *Proceedings of the 2021 Conference of the NAACL: Human Language Technologies*, pages 3002–3017, Online, June 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.naacl-main.239. URL <https://aclanthology.org/2021.naacl-main.239>.
- Wenhu Chen, Jianshu Chen, Pengda Qin, Xifeng Yan, and William Yang Wang. Semantically conditioned dialog response generation via hierarchical disentangled self-attention. In Anna Korhonen, David Traum, and Lluís Màrquez, editors, *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3696–3709, Florence, Italy, July 2019. Association for Computational Linguistics. doi: 10.18653/v1/P19-1360. URL <https://aclanthology.org/P19-1360>.
- Lei Cui, Shaohan Huang, Furu Wei, Chuanqi Tan, Chaoqun Duan, and Ming Zhou. SuperAgent: A customer service chatbot for E-commerce websites. In Mohit Bansal and Heng Ji, editors, *Proceedings of ACL 2017, System Demonstrations*, Vancouver, Canada, July 2017. Association for Computational Linguistics. URL <https://aclanthology.org/P17-4017>.
- Ashley Lewis and Michael White. Mitigating harms of llms via knowledge distillation for a virtual museum tour guide. In *Proceedings of the 1st Workshop on Taming Large Language Models: Controllability in the era of Interactive Assistants!*, 2023.
- Zekun Li, Shinda Huang, Jiangtian Wang, Nathan Zhang, Antonis Antoniadis, Wenyue Hua, Kaijie Zhu, Sirui Zeng, Chi Wang, William Yang Wang, and Xifeng Yan. Sopbench: Evaluating language agents at following standard operating procedures and constraints, 2025. URL <https://arxiv.org/abs/2503.08669>.
- Do June Min, Paloma Sodhi, and Ramya Ramakrishnan. Workflow-guided response generation for task-oriented dialogue, 2023. URL <https://arxiv.org/abs/2311.08300>.
- Johannes EM Mosig, Shikib Mehri, and Thomas Kober. Star: A schema-guided dialog dataset for transfer learning. *arXiv preprint arXiv:2010.11853*, 2020.
- Subhrangshu Nandi, Arghya Datta, Nikhil Vichare, Indranil Bhattacharya, Huzefa Raja, Jing Xu, Shayan Ray, Giuseppe Carenini, Abhi Srivastava, Aaron Chan, Man Ho Woo, Amar Kandola, Brandon Theresa, and Francesco Carbone. Sop-bench: Complex industrial sops for evaluating llm agents, 2025. URL <https://arxiv.org/abs/2506.08119>.
- Shashi Narayan, Joshua Maynez, Reinald Kim Amplayo, Kuzman Ganchev, Annie Louis, Fantine Huot, Anders Sandholm, Dipanjan Das, and Mirella Lapata. Conditional generation with a question-answering blueprint. *Transactions of the Association for Computational Linguistics*, 11:974–996, 2023. doi: 10.1162/tacl_a_00583. URL <https://aclanthology.org/2023.tacl-1.55>.
- Stefania Raimondo, Christopher Pal, Xiaotian Liu, David Vazquez, and Hector Palacios. Improving generalization in task-oriented dialogues with workflows and action plans. *arXiv preprint arXiv:2306.01729*, 2023.
- Abhinav Rastogi, Xiaoxue Zang, Srinivas Sunkara, Raghav Gupta, and Pranav Khaitan. Towards scalable multi-domain conversational agents: The schema-guided dialogue dataset. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, 2020.
- Shamik Roy, Sailik Sengupta, Daniele Bonadiman, Saab Mansour, and Arshit Gupta. FLAP: Flow-adhering planning with constrained decoding in LLMs. In Kevin Duh, Helena Gomez, and Steven Bethard, editors, *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1:*